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NCRB 2021 | CRIME MOTIVE ANALYTICS

Data Analysis, Machine Learning and Intelligence Dashboard

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ABSTRACT

This project presents an analytical study of National Crime Records Bureau (NCRB) 2021 data on recorded crime motives across Indian States and Union Territories. Python-based data processing and visualization are used to examine national motive distribution, compare regional counts, study motive composition, and identify concentration and association patterns. The project also demonstrates exploratory classification, clustering, Principal Component Analysis (PCA), and an integrated dashboard to summarize the findings.

Keywords: NCRB 2021, Crime Motives, Exploratory Data Analysis, Python, Machine Learning, Clustering, PCA, Dashboard.

INTRODUCTION

Crime is one of the major challenges affecting society, and understanding the reasons behind criminal activities is important for developing better awareness and preventive strategies. However, raw crime statistics contain large amounts of data that are difficult to interpret and compare manually. The main objective of this project, **NCRB 2021 – Crime Motive Analytics**, is to transform crime-related data published by the National Crime Records Bureau (NCRB) into meaningful information using data analytics and machine learning techniques.

This project focuses on analysing recorded crime motives across different Indian States and Union Territories. It identifies the most frequently recorded crime motives, compares regional crime statistics, examines relationships between different motives and explores similarities among regions. Various techniques, including Exploratory Data Analysis (EDA), correlation analysis, classification, clustering and Principal Component Analysis (PCA), are implemented using Python. An interactive-style analytical dashboard is also generated to present the major findings visually.

The ultimate aim of this project is to make complex crime statistics easier to understand and support data-driven insights that may contribute to better social awareness and informed decision-making.

METHODOLOGY

The dataset is processed in Python using Pandas and NumPy. Data preparation includes standardizing fields, selecting State/UT records, converting numeric values, and preparing motive-wise and region-wise summaries. Matplotlib and Seaborn are used for visualizations. The analytical workflow includes national distribution, State/UT comparison, regional heatmap, Pareto analysis, correlation and association analysis, exploratory classification, K-Means and hierarchical clustering, PCA, and a dashboard.

TOOLS USED

Python 3.13, Anaconda Spyder, Pandas, NumPy, Matplotlib, Seaborn and Scikit-learn.

RESULTS AND ANALYSIS

1. National Motive Distribution

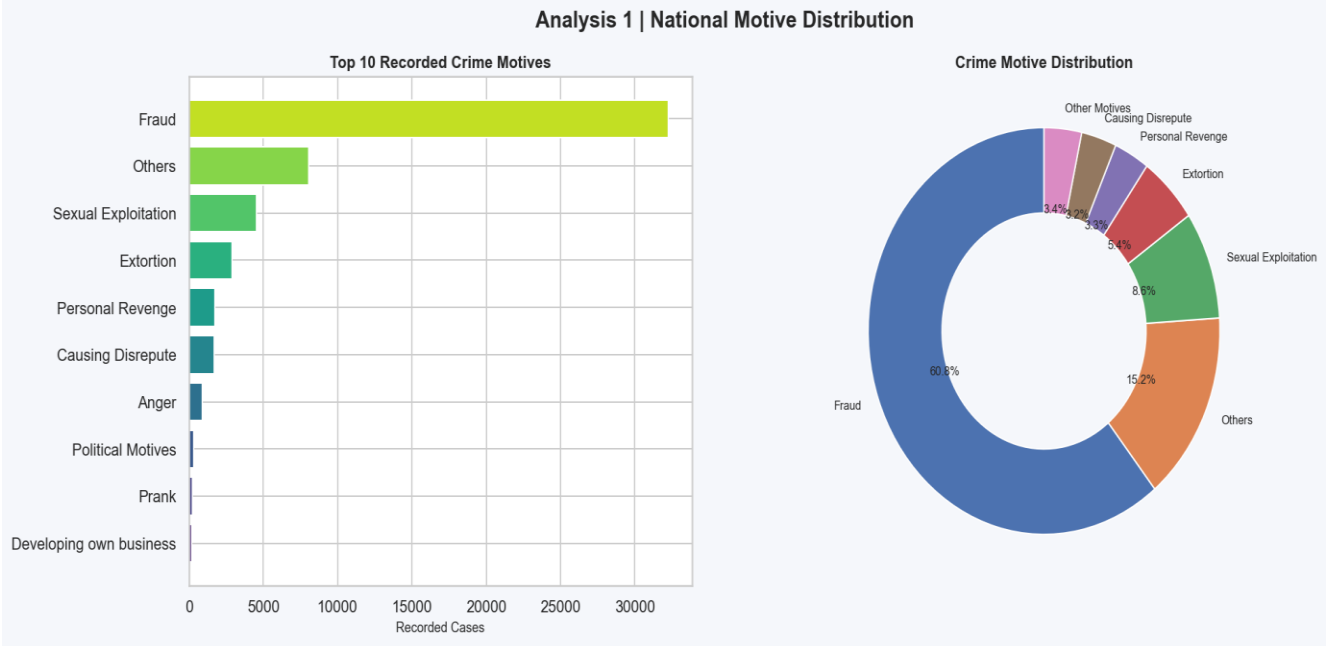


Fig. 1: National motive distribution and top ten recorded crime motives.

Fraud is the largest recorded motive category, accounting for approximately 60.8% of the total shown in the chart. The 'Others' category contributes 15.2%, followed by Sexual Exploitation at 8.6% and Extortion at 5.4%. The remaining displayed categories each contribute a comparatively small share. This indicates that the national total is strongly concentrated in the Fraud category.

2. State/UT Comparison

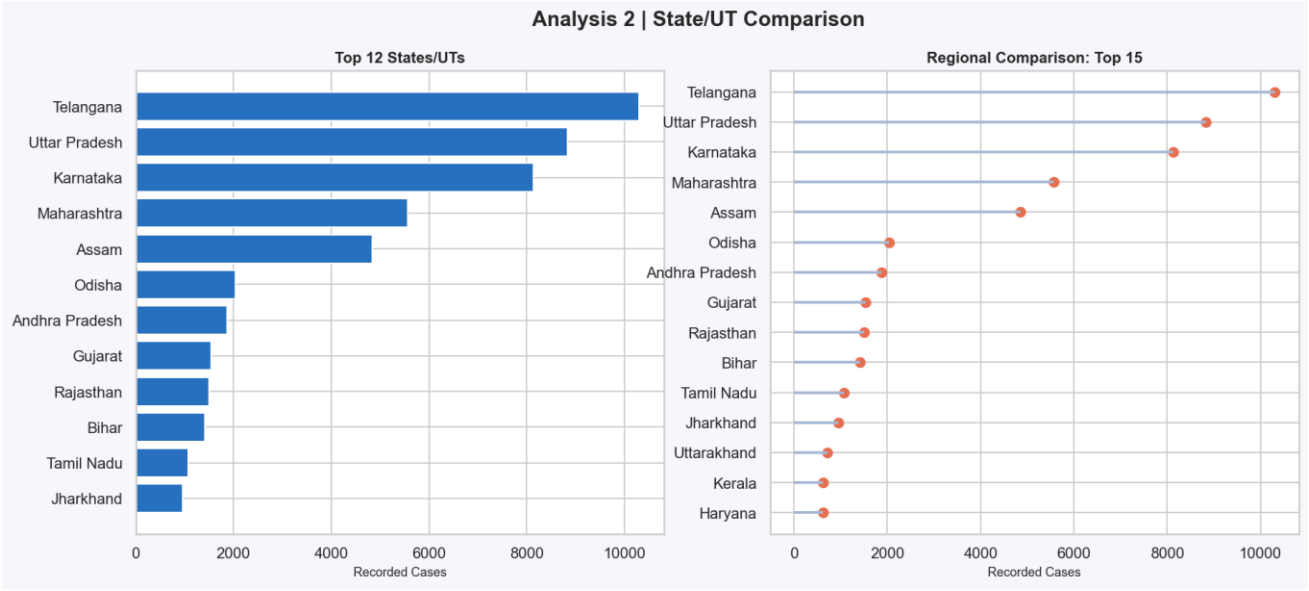


Fig. 2: Top State/UT recorded counts and regional comparison.

Telangana has the highest recorded total among the regions shown, followed by Uttar Pradesh and Karnataka. Maharashtra and Assam are the next major contributors in the displayed ranking. The comparison highlights differences in recorded counts across States/UTs; these totals should not be interpreted as population-adjusted crime rates.

3. Regional Motive Composition Heatmap

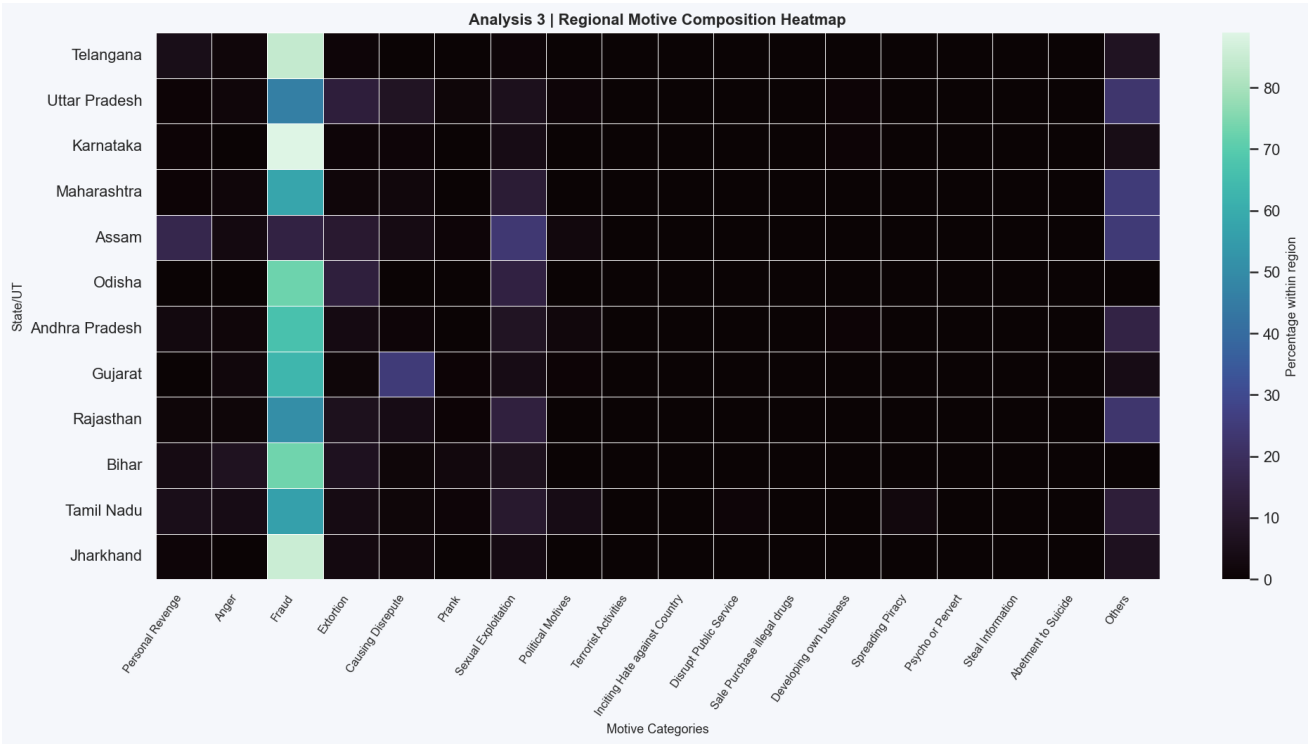


Fig. 3: Motive composition by leading States/UTs, expressed as percentage within each region.

The heatmap shows that Fraud forms a prominent share in several leading regions, including Telangana, Karnataka, Bihar and Jharkhand. The relative contribution of other motives differs by region. The visualization therefore indicates that the composition of recorded motives is not identical across States/UTs.

RESULTS AND ANALYSIS (CONTINUED)

4. Pareto Concentration Analysis

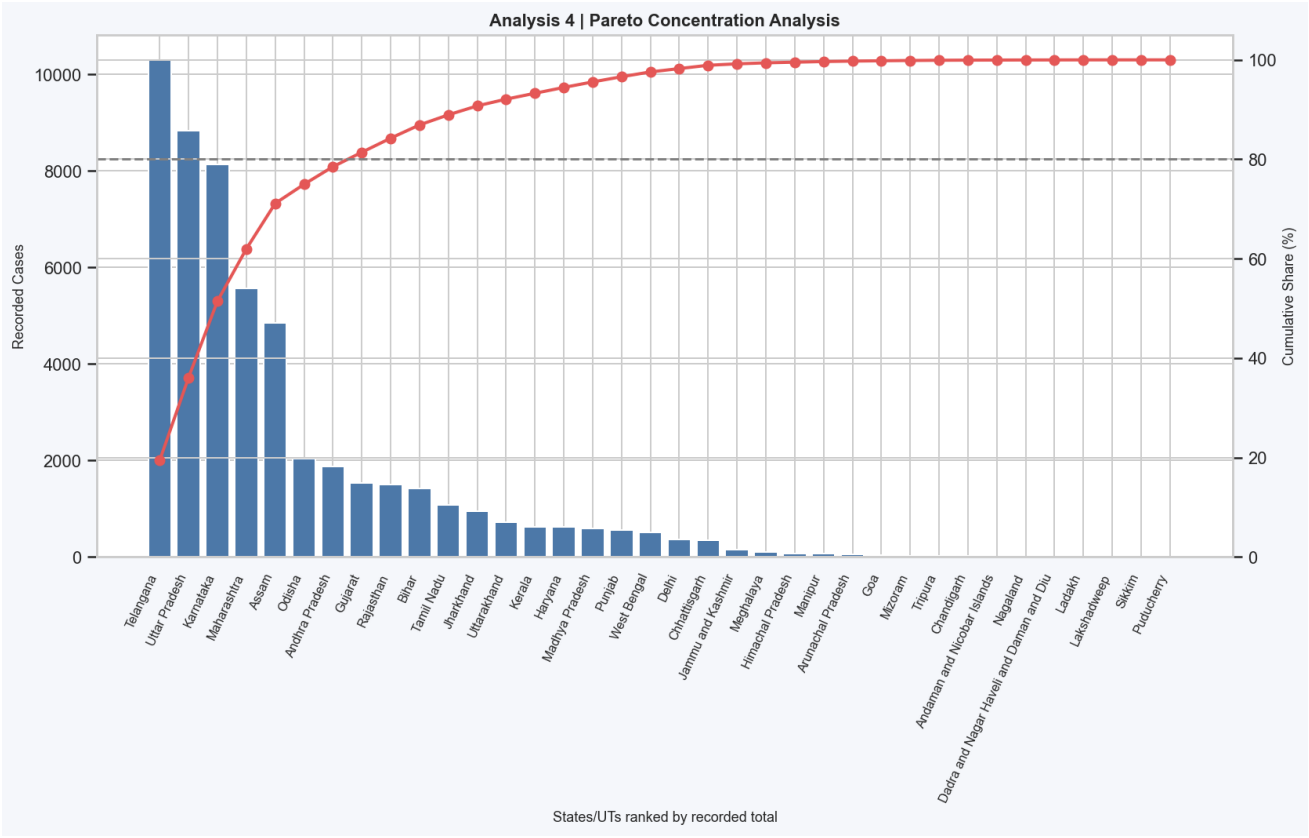


Fig. 4: State/UT totals ranked in descending order with cumulative percentage.

The Pareto chart ranks States/UTs by recorded totals and plots their cumulative share. The cumulative curve rises rapidly for the first few regions, showing that a relatively small group of States/UTs contributes a large portion of the total recorded cases. The dashed 80% reference line helps identify the point at which the cumulative contribution crosses that threshold.

5. Motive Relationships and Correlation

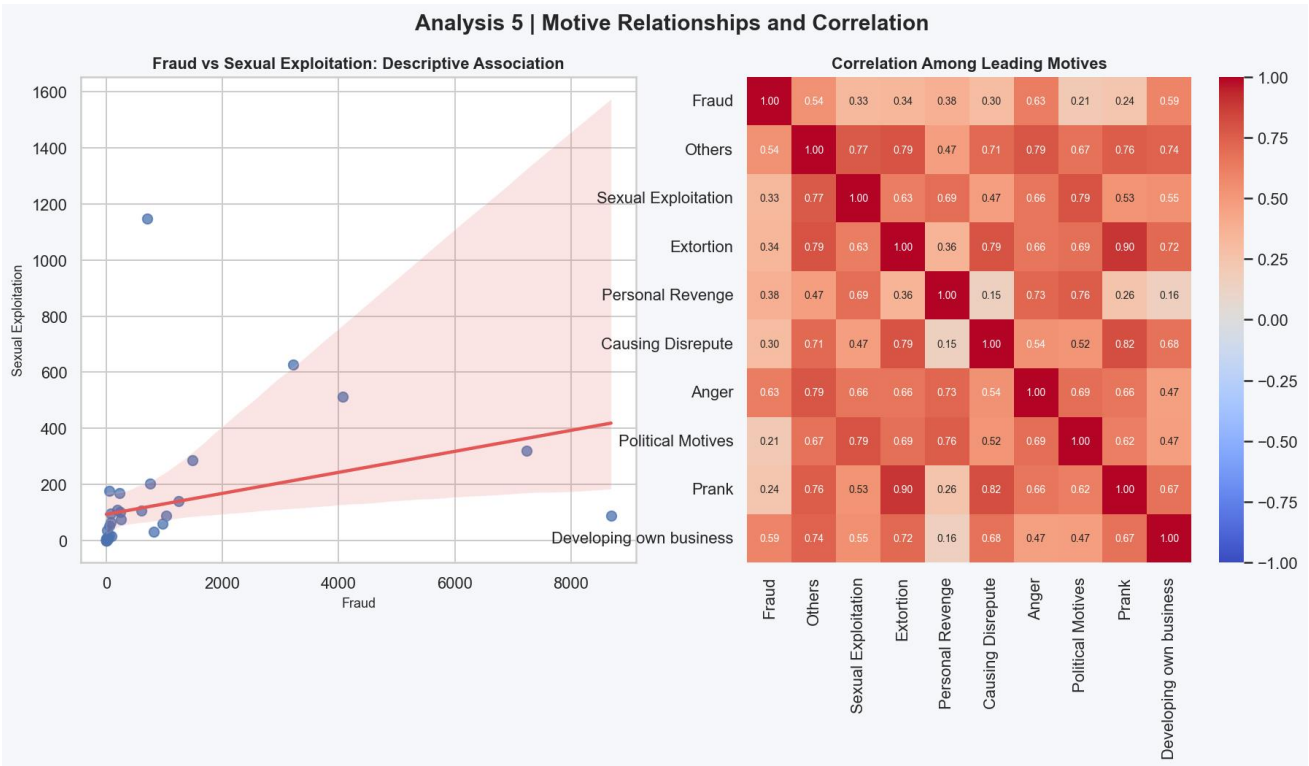


Fig. 5: Fraud and Sexual Exploitation association plot and correlation matrix of leading motives.

The correlation matrix shows positive correlations among several leading motive counts across regions. For example, Extortion and Prank show a high displayed correlation of 0.90, while Others and Sexual Exploitation show 0.77. These are descriptive relationships between aggregated counts and do not establish that one motive causes another. The scatter plot also shows substantial variation between regions.

6. Classification Diagnostics

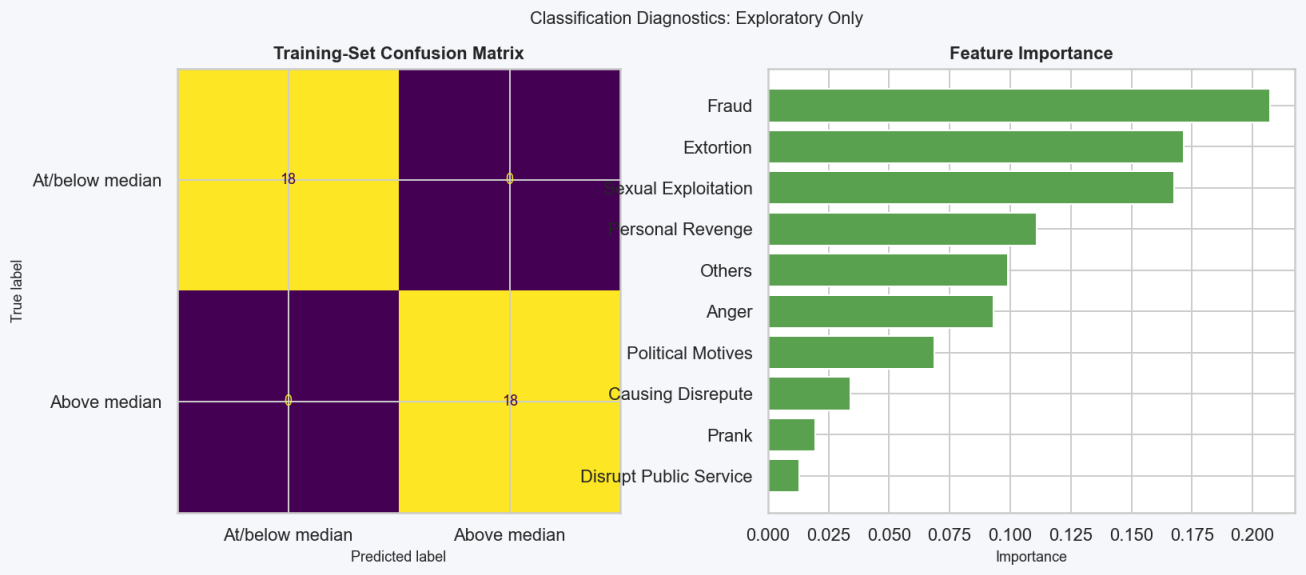


Fig. 6: Training-set confusion matrix and feature-importance ranking.

The displayed training-set confusion matrix contains 18 correct predictions in each of the two classes and no off-diagonal errors for this training view. Fraud has the highest feature-importance value, followed by Extortion and Sexual Exploitation. Because the figure is explicitly labelled as a training-set diagnostic and exploratory, these results should not be presented as evidence of generalization to unseen data.

7. Clustering Analysis

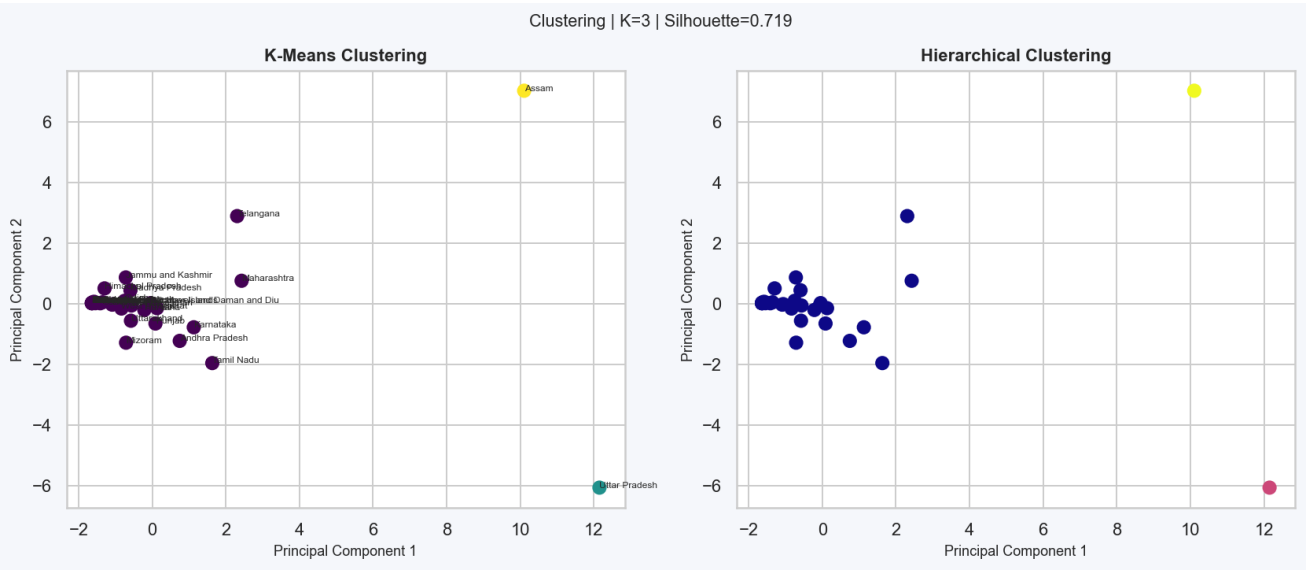


Fig. 7: K-Means and hierarchical clustering visualized in principle-component space.

The K-Means algorithm successfully grouped 36 States/UTs into three clusters, achieving a Silhouette Score of 0.719, indicating good cluster separation. Most regions show similar crime-motive patterns, while Assam and Uttar Pradesh appear distinctly separated. Hierarchical clustering further highlights regional similarities and differences. Overall, the analysis identifies regions with similar and distinct crime-motive patterns.

8. Principal Component Analysis

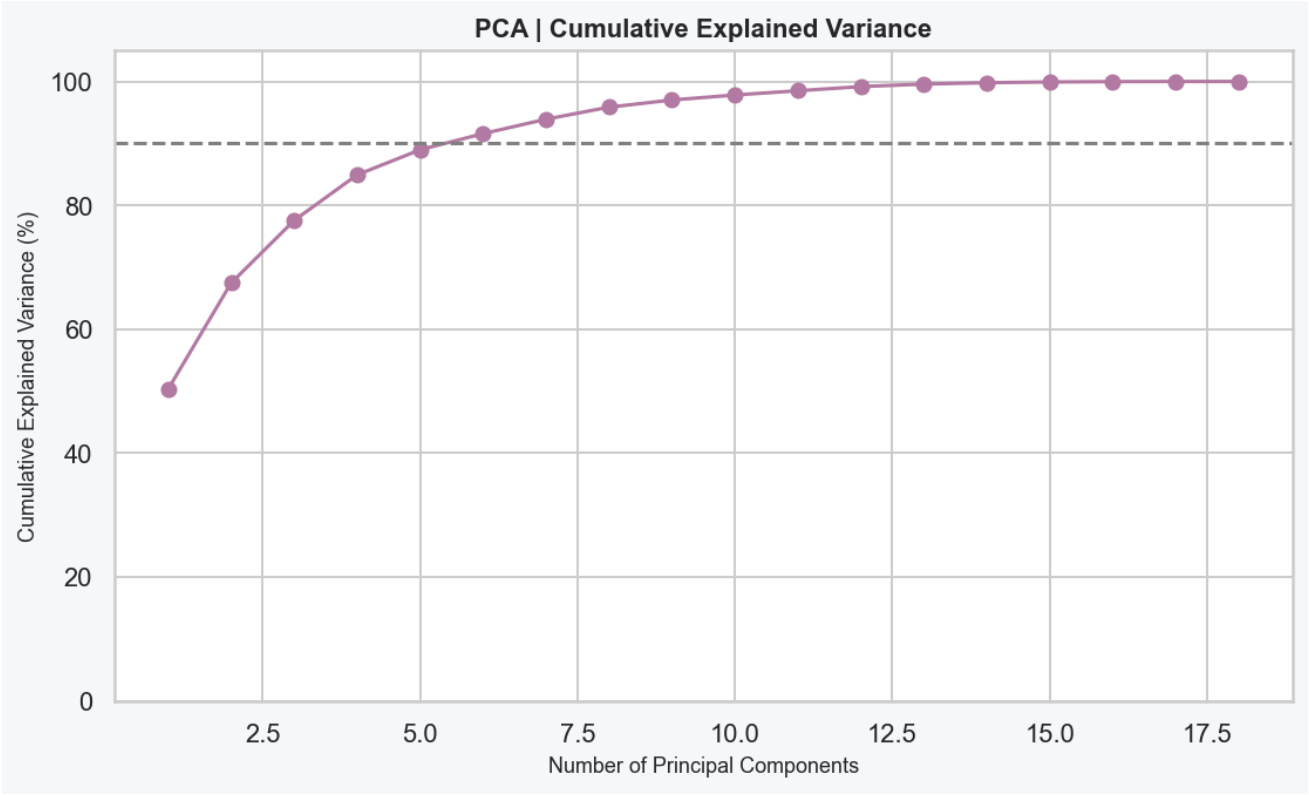


Fig. 8: Cumulative explained variance across principal components.

The cumulative explained-variance curve increases as additional principal components are included. The plotted curve reaches approximately 90% around the sixth component and continues toward 100% as more components are added. This indicates that a smaller number of components can summarize much of the variation in the standardized motive data.

9. Integrated Crime Motive Dashboard



Fig. 9: NCRB 2021 Crime Motive Intelligence Dashboard.

The dashboard summarizes 52,974 recorded cases, 36 regions and 18 motive categories. It combines leading-motive and highest-count-region charts, motive composition, regional concentration, association-rule results, PCA variance and exploratory classification scores. It provides a consolidated view of the analysis in one interface.

CONCLUSION

The project demonstrates how Python-based analytics can convert NCRB 2021 aggregated crime-motive records into meaningful visual summaries. The analysis identifies Fraud as the dominant recorded motive in the national distribution and Telangana as the highest-count region in the displayed State/UT comparison. The heatmap and correlation analysis reveal variation and positive associations among motive counts across regions, while Pareto analysis highlights concentration in a smaller set of regions. Clustering and PCA provide exploratory views of similarities and variation in regional motive profiles. The integrated dashboard brings these results together for easier interpretation. Since the source data are one-year aggregate counts, the findings are descriptive and should not be interpreted as causal explanations, population-adjusted risk measures or validated predictions.

FUTURE SCOPE

- Extend the analysis using multiple years of NCRB data to examine trends over time.
- Use population-adjusted rates and additional socioeconomic indicators for fairer regional comparisons.
- Evaluate machine-learning models using a suitable hold-out test set and cross-validation.
- Develop interactive filters and drill-down views for State/UT and motive-level exploration.

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